An Explainable Machine Learning Framework for Early Multi-Disease Risk Prediction Using Healthcare Data Analytics

A Mini Project Report Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

The mini-project report entitled **"An Explainable Machine Learning Framework for Early Multi-Disease Risk Prediction Using Healthcare Data Analytics"** is a record of bona fide work carried out by [Names and Roll Numbers] in partial fulfillment of the requirements for the award of the B.Tech degree at K L (Deemed to be) University. The work presented in this report is original to the best of our knowledge, and all datasets, tools, code, and external sources used have been appropriately acknowledged.

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CERTIFICATE

This is to certify that the mini-project report entitled **"An Explainable Machine Learning Framework for Early Multi-Disease Risk Prediction Using Healthcare Data Analytics"** is a bonafide work carried out by [E. Supraja - 2420030357, M. Varshitha - 2420030479, C. Keerthana - 2420030581, CH. Bhargavi - 2420030588] under the supervision of [Dr. Katanguri Swapnika] during the academic year [2025–2026], in partial fulfillment of the requirements for the award of the Bachelor of Technology degree at K L (Deemed to be) University.

Signature of the Project Supervisor

Signature of the HOD **Signature of the External Examiner**

ACKNOWLEDGEMENT

The success in this project would not have been possible but for the timely help and guidance rendered by many people. Our wish to express my sincere thanks to all those who has assisted us in one way or the other for the completion of my project.

Our greatest appreciation to my guide, **Dr. Katanguri Swapnika**, Department of Computer Science which cannot be expressed in words for his/her tremendous support, encouragement and guidance for this project.

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We thank all the members of teaching and non-teaching staff members, and also who have assisted me directly or indirectly for successful completion of this project. Finally, I sincerely thank my parents, friends and classmates for their kind help and cooperation during my work.

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1. ABSTRACT

The increasing availability of healthcare data creates opportunities to identify potential disease risks before severe complications develop. However, interpreting multiple clinical indicators and understanding machine-learning predictions can be challenging. This project proposes EarlyMed-AI, a machine-learning-based healthcare analytics framework for early multi-disease risk prediction and explainable clinical decision support.

The framework combines disease-specific prediction models with a web-based application that accepts structured clinical inputs and supports medical-report processing. The machine-learning models considered in the project include Logistic Regression, Decision Tree, Random Forest, XGBoost, and Gaussian Naive Bayes. The system incorporates data preprocessing, missing-value handling, model evaluation, risk estimation, and explainability using SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations).

A Flask-based interface presents prediction results, model-performance information, and local explanations to improve the interpretability of the output. A composite Health Risk Index is also presented as a research-oriented summary of risk estimates. The framework is intended to support preventive healthcare analysis and help users understand the factors influencing model predictions. It is not a replacement for professional diagnosis or treatment.

Team Name: Early Med

Team Members and Responsibilities:

1 [Edigi Supraja,2420030357 – software testing, manuscript preparation, experimentation,support, result verification, and quality review

2 [Mandha Varshitha, 2420030479 – application development, user-interface integration, application workflow design, documentation, and integration of prediction and reporting components]

3 [Chenna Keerthana, 2420030581 – literature review, results organization, explainability analysis, interpretation of SHAP and LIME outputs, and preparation of comparative analysis]

4 [Chintalapati Bhargavi Devi, 2420030588 – project implementation, data and model pipeline development, preprocessing integration, model evaluation workflow, and technical implementation documentation.]

2.INTRODUCTION

2.1 Background

Healthcare systems generate large volumes of clinical data, including demographic information, laboratory measurements, medical histories, and diagnostic reports. These data can contain patterns associated with disease risk. Traditional clinical assessment relies on medical expertise, laboratory investigations, and interpretation of multiple indicators. Machine-learning techniques can complement this process by identifying statistical relationships between clinical features and known outcomes.

Early disease-risk prediction is particularly relevant for conditions such as diabetes, cardiovascular disease, and chronic kidney disease (CKD). These conditions can involve several interacting risk factors, and early identification of potential risk may help motivate timely clinical assessment.

However, a prediction alone is not always sufficient. Users may also need to understand why a model assigned a particular risk estimate. This motivates the integration of Explainable AI techniques with predictive healthcare analytics.

2.2 Motivation

The main motivation for developing EarlyMed-AI is to combine three capabilities in one framework:

1. Predictive analytics: Estimate disease risk from available clinical indicators.
2. Explainable analytics: Identify features contributing to individual predictions.
3. Clinical decision support: Present predictions and explanations through a comprehensible web interface.

The project also explores the comparison of several machine-learning algorithms rather than assuming that a single algorithm performs best for every disease.

2.3 Proposed system

EarlyMed-AI uses a modular architecture consisting of data input, preprocessing, disease-specific prediction, model evaluation, explainability, risk presentation, and a Flask-based application interface.

The system is designed to accept supported structured clinical inputs and medical reports. When a report is uploaded, the application can attempt to extract relevant information, subject to the extraction and validation capabilities implemented in the software.

2.4 Target users

The intended users include:

- Researchers studying machine learning in healthcare.
- Students developing healthcare analytics applications.
- Healthcare professionals evaluating decision-support prototypes.
- Users seeking an understandable presentation of model-generated risk estimates.

The prototype should be treated as a research and educational system until it has undergone appropriate clinical validation.

2.5 Scope

The project focuses on structured clinical data analysis, multiple disease-risk prediction, comparison of classical machine-learning models, explainability, and web-based result presentation. It does not independently confirm a diagnosis, prescribe medication, or replace medical consultation.

3. LITERATURE SURVEY

Your lecturer specifically wants recent published research, comparisons with existing literature, and a clearly identified research gap. The literature survey should distinguish published methods from the baseline algorithms implemented in your own project.

3.1 Research areas to review

Research area	What to investigate	Limitation or gap to examine
ML-based diabetes prediction	Clinical datasets, classification algorithms, validation strategy	Whether the model generalizes beyond its dataset
Cardiovascular risk prediction	Clinical variables, risk classification, calibration	Whether predicted probabilities are reliable
CKD prediction	Laboratory features, missing-data handling, classification	Whether missing or abnormal measurements are handled appropriately
Explainable healthcare AI	SHAP, LIME, feature attribution	Whether explanations are stable and understandable
Multi-disease prediction	Shared interfaces or multiple disease-specific models	Whether models and outputs are evaluated separately

3.2 Research-gap statement

Many healthcare machine-learning studies concentrate on predicting one disease or comparing predictive algorithms on a particular dataset. Other studies focus on explaining model predictions without integrating the complete prediction and explanation workflow into an accessible application.

The proposed EarlyMed-AI framework aims to integrate disease-specific prediction, comparative model evaluation, SHAP and LIME explanations, report processing, and dashboard-based presentation within one modular prototype. Its potential contribution is the integration of these components and the systematic evaluation of their performance, stability, and computational cost.

This is a proposed integration-based contribution, not a claim that no existing system has similar features. The final literature review must establish the novelty by comparing EarlyMed-AI against recent published systems.

3.3 Literature-survey table for your report

Use the following structure and fill it with verified papers published from 2021 to 2026.

Author and year	Disease/task	Dataset	Method	Reported metrics	Limitation identified
Paper 1	Diabetes prediction	As reported in paper	Published method	Actual reported results	Based on paper
Paper 2	Cardiovascular prediction	As reported	Published method	Actual reported	Based on paper

		in paper		results	
Paper 3	CKD prediction	As reported in paper	Published method	Actual reported results	Based on paper
Paper 4	Explainable healthcare ML	As reported in paper	SHAP/LIME or other method	Actual reported results	Based on paper
Paper 5	Multi-disease prediction	As reported in paper	Published framework	Actual reported results	Based on paper

Do not invent author names, datasets, DOI numbers, accuracy values, or limitations. These rows are a template, not citations to real papers. Each row should be completed from a paper you have actually read.

4. PROBLEM STATEMENT AND OBJECTIVES

Problem Statement:

Healthcare datasets contain valuable information for estimating disease risk, but developing a system that combines multiple disease-specific predictions with understandable explanations and accessible result visualization remains a practical engineering challenge.

The project addresses this challenge by developing a machine-learning-based framework that predicts potential disease risks from supported clinical inputs and presents interpretable information to support further assessment.

Problem Description:

Different diseases require different clinical features and predictive models. Missing data, differences in feature distributions, class imbalance, and differences between training and deployment data can affect prediction quality. Furthermore, a model's output may be difficult to interpret when the system provides only a risk estimate without explaining which input features influenced it.

EarlyMed-AI addresses these concerns through disease-specific preprocessing, comparative machine-learning evaluation, explainability methods, and a web interface that presents the results in a consistent format.

Objectives:

Develop a modular healthcare analytics framework for multiple disease-risk prediction tasks.

Identify and preprocess the clinical features required for each supported disease.

Train and compare Logistic Regression, Decision Tree, Random Forest, XGBoost, and Gaussian Naive Bayes where suitable.

Evaluate models using accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC, as applicable.
Generate local feature explanations using SHAP and LIME where supported.
Present predictions, explanations, and model-performance information through a Flask-based interface.
Investigate prediction stability, computational execution time, model storage requirements, and scalability. Discuss the limitations, privacy considerations, and responsible use of healthcare predictions.
Discuss the limitations, privacy considerations, and responsible use of healthcare predictions.

4.4 Expected outcomes

The expected outcome is an integrated research prototype that demonstrates disease-specific predictions, comparative model evaluation, explanation visualizations, and dashboard presentation. Actual predictive performance and usability must be established through experiments.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

5.1 Hardware requirements

Component	Suggested configuration
Processor	Modern multi-core CPU
RAM	8 GB minimum recommended; 16 GB may help with larger datasets.
Storage	Sufficient space for datasets, model files, dependencies, and results
GPU	Optional for the classical ML models listed
Display	Suitable for viewing the dashboard and research plots
Internet	Required for initial package installation and dataset retrieval when applicable

These are suggested configurations, not measurements of the hardware used during your experiments

Requirements:

• Python 3.x with Jupyter Notebook or Google Colab.

• Libraries as required: pandas, NumPy, scikit-learn, Matplotlib/Seaborn, NLTK or spaCy.

• Dataset in CSV/TSV/JSON or another documented format; specify the exact source and licence.

• Optional: Streamlit/Flask/FastAPI for a demonstration interface; Git for version control.

5.2 Software requirements

Software/tool	Purpose
Python 3.x	Main programming language
pandas	Dataset loading and manipulation
NumPy	Numerical computation
scikit-learn	Preprocessing, model training, and evaluation
XGBoost	Gradient-boosted tree classification
SHAP	Feature-attribution explanations
LIME	Local surrogate explanations
Matplotlib	Plots and result visualization
Flask	Web application
SQLite	Local structured record storage, if used by the implementation

6. IMPLEMENTATION

6.1 System architecture

The implementation can be described as the following sequence:

1. Data input

Clinical datasets, structured inputs, supported medical reports

2. Data extraction and preprocessing

Validation, missing-value handling, feature alignment, encoding and scaling where required

3. Disease-specific ML models

Logistic Regression, Decision Tree, Random Forest, XGBoost and Gaussian Naive Bayes, where applicable

4. Model evaluation and selection

Held-out evaluation, classification metrics and disease-specific model selection

5. Explainability

SHAP feature attribution and LIME local explanations

6. Result presentation

Disease-risk estimates, Health Risk Index, charts and Flask dashboard

6.2 Data collection and input handling

The framework supports disease-specific data rather than assuming that every disease can be predicted from the same feature set.
The diabetes prediction task may use features such as glucose, BMI, age, blood pressure, insulin, and other dataset-supported variables. Cardiovascular prediction may use its own set of clinical and demographic features, while CKD prediction may require renal-functionmeasurements and related clinical indicators.

The exact input variables must match the datasets and trained models in the implementation. Features must not be invented or supplied in an incorrect order.

For report uploads, the application may extract values from a supported PDF or image, validate the extracted fields, and map them to the appropriate model inputs. Missing or ambiguous fields must be handled explicitly rather than silently interpreted as normal values.

6.3 Exploratory data analysis (EDA)

EDA is used to understand the dataset before model training. It should include the following analyses.

Dataset structure: Report the number of records, feature count, target variable, data types, duplicate records, and missing values.

Target distribution: Count the positive and negative cases for each disease. This reveals class imbalance that may make accuracy alone misleading.

Feature distributions: Use histograms and box plots to inspect measurements such as glucose, BMI, age, blood pressure, and renal indicators. Examine outliers and unusual values without automatically assuming that every outlier is an error.

Feature relationships: Correlation matrices and pairwise plots can help identify relationships among numerical features. Correlation does not establish causation and should not be used alone to remove clinically relevant variables.

Missing-data analysis: Calculate the number and percentage of missing values in each feature. Decide on an imputation method based on the training data and the feature's meaning.

Data leakage prevention: Split the data into training and test sets before fitting imputers, scalers, encoders, feature selectors, or other learned transformations. Fit these operations using training data only, then apply the fitted transformations to validation and test data.

Suggested EDA table:

EDA item	Information to report
Dataset size	Actual number of rows and columns
Target classes	Positive and negative counts

Missing values	Count and percentage per feature
Duplicate records	Number found and treatment
Feature statistics	Mean, standard deviation, median, range
Outlier analysis	Features examined and method used
Class imbalance	Class proportions and any mitigation
Correlation analysis	Strong relationships and interpretation

6.4 Data preprocessing

The preprocessing stage should contain only operations relevant to each dataset.

1. Validate the expected columns and data types.
2. Remove duplicates only where appropriate and documented.
3. Handle missing values using an explicitly defined strategy.
4. Encode categorical features if the selected model requires numerical input.
5. Apply scaling where appropriate, especially for models sensitive to feature scale.
6. Align the feature names and order with the model's training pipeline.
7. Separate training, validation, and test data appropriately.

For example, scaling is usually more important for Logistic Regression than for tree-based models. Preprocessing choices should therefore be justified instead of automatically applying every transformation to every model.

6.5 Machine-learning models

Model	Main principle	Advantages	Limitations
Logistic Regression	Estimates class probability using a logistic function	Efficient and relatively interpretable	May not capture complex nonlinear relationships without feature engineering

Decision Tree	Splits feature space using decision rules	Easy to visualize and explain	Can overfit
Random Forest	Combines predictions from multiple decision trees	Often handles nonlinear interactions well	Larger model and reduced direct interpretability
XGBoost	Builds an additive ensemble of boosted trees	Can model complex feature relationships	Requires careful tuning and leakage-free evaluation
Gaussian Naive Bayes	Uses Bayes' theorem with conditional Gaussian feature assumptions	Fast and computationally light	Independence and distribution assumptions may be unrealistic

The best algorithm must be selected from actual held-out results, not from general expectations about which model is usually accurate.

6.6 Mathematical formulation

Let a clinical record be represented by a feature vector:

$$x = [x_1, x_2, ..., x_d]^T$$

where d is the number of model input features.

For binary classification, the target label is $y \in \{0, 1\}$.

Logistic Regression

$$P(y = 1 \mid x) = 1 / 1 + \exp [-(\beta_0 + \beta^T x)]$$

Here, β_0 is the intercept and β contains the learned feature coefficients.

Random Forest

For M decision trees, the average estimated positive-class probability can be expressed as

$$\widehat{p}\text{RF}(y = 1 \mid x) = \frac{1}{M} \Sigma_{m=1}^{M} \widehat{p}_m(y = 1 \mid x)$$

The final classification is determined by applying the chosen threshold to the estimated probability.

XGBoost

The additive boosted-tree prediction can be written as

$$\widehat{y}_i = \Sigma_{k=1}^{K} f_k(x_i), \quad f_k \in F$$

where f_k is a learned tree and F is the set of permitted tree functions. For binary classification, the resulting score is mapped to a probability using a logistic transformation.

Gaussian Naive Bayes

$$P(y = c \mid x) \propto P(y = c) \prod_{j=1}^{d} P(x_j \mid y = c)$$

For a continuous feature x_j, the Gaussian class-conditional likelihood is

$$P(x_j \mid y = c) = \frac{1}{\sqrt{2\pi\sigma_{jc}^2}} \exp\left[-\frac{(x_j - \mu_{jc})^2}{2\sigma_{jc}^2}\right]$$

where μ_{jc} and σ_{jc}^2 are estimated from training data.

Hybridization and model selection

If your implemented system selects the best model for each disease using held-out ROC-AUC, write:

$$m_d^* = \arg\max_{m \in \mathcal{M}_d} \text{AUC}_{\text{val}}(m, d)$$

where M_d is the candidate model set for disease d.

This is model selection, not mathematical hybridization of model predictions. Do not call the system a hybrid ensemble unless it actually combines multiple models into a single

prediction.

6.7 Explainable AI

SHAP attributes a model prediction to individual input features. Its additive representation is

$$f(x) = \phi_0 + \sum_{j=1}^{d} \phi_j$$

where ϕ_0 is the baseline output and ϕ_j is the contribution attributed to feature j, on the explainer's output scale.

LIME approximates the model locally using an interpretable surrogate:

$$\boldsymbol{\xi(\mathbf{x}) = \arg\min_{g \in \mathcal{G}} [\mathcal{L}(f, g, \pi_{\mathbf{x}}) + \Omega(g)]}$$

Here, f is the original model, g is the local surrogate, π_x weights samples by proximity to the instance, and $\Omega(g)$ penalizes complexity.

SHAP and LIME help explain model behavior; they do not prove that a feature causes a disease. Explanation results also depend on the model, background data, perturbation strategy, and feature preprocessing.

6.8 Performance evaluation formulas

Let TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives.

Accuracy

$$Accuracy = \frac{TP+TN}{TP+TN+FP+FN}$$

Precision

$$Precision = \frac{TP}{TP+FP}$$

Recall / Sensitivity

$$Recall = \frac{TP}{TP+FN}$$

Specificity

$$Specificity = \frac{TN}{TN+FP}$$

F1-score

$$F_1 = 2\frac{\mathbf{Precision} \times \mathbf{Recall}}{\mathbf{Precision} + \mathbf{Recall}}$$

ROC-AUC: The area under the receiver operating characteristic curve, which plots sensitivity against false-positive rate across decision thresholds.

$$FPR = \frac{FP}{FP+TN}$$

PR-AUC: The area under the precision-recall curve. It is particularly informative when positive cases are relatively uncommon.

For all metrics, specify the positive class, evaluation split, threshold-selection procedure, and averaging method where applicable. Handle zero denominators using the metric implementation's documented behavior.

6.9 Stability, execution time, storage, and scalability

Your lecturer asked for these additional evaluation dimensions, so they should be included in the implementation and results.

Model stability: Train the same model under multiple predefined random seeds or stratified splits. Report the mean and standard deviation of a selected metric.

$$\bar{s} = \frac{1}{R}\sum_{r=1}^{R} s_r$$

$$\sigma_s = \sqrt{\frac{1}{R}\sum_{r=1}^{R}(s_r - \bar{s})^2}$$

Here s_r is the score from run r, and R is the number of runs. A lower standard deviation indicates less observed variation under the tested conditions, but it does not establish universal robustness.

Execution time: Measure training time and prediction latency separately, using the same hardware, software environment, and test workload.

$$T_{avg} = \frac{1}{N}\sum_{i=1}^{N} t_i$$

Here t_i is the measured prediction time for request i. Report units, sample count, and whether report parsing and explanation generation are included.

Model storage size: Measure the saved model artifact in bytes or megabytes. Include the preprocessing pipeline if it is stored with the model.

Scalability: Test the application with increasing batch sizes or concurrent requests. Record throughput, latency, memory use, and error rate. A theoretical statement that the system is scalable is not a substitute for measurements.

6.10 Implementation modules

The report can organize the software into these modules, matching the files that actually exist in your project:

- Data loading and validation
- Exploratory data analysis
- Preprocessing and feature alignment
- Model training and model selection
- Prediction and risk calculation
- SHAP and LIME explanations
- Report extraction and validation
- Flask routes and dashboard
- Result storage and history, if implemented
- Evaluation plots and experiment reporting

7. RESULTS AND DISCUSSION

This section should contain evidence from the actual EarlyMed-AI implementation. Your screenshots can demonstrate the interface and explainability features, but they cannot by themselves establish general predictive performance.

7.1 Dataset summary

Create a separate dataset-summary table for each disease.

Dataset property	Diabetes	Cardiovascular disease	CKD
Number of records	Measure	Measure	Measure
Number of input features	Measure	Measure	Measure
Positive cases	Measure	Measure	Measure
Negative cases	Measure	Measure	Measure
Missing values	Measure	Measure	Measure
Train/test split	Report actual split	Report actual split	Report actual split

Replace “Measure” with values calculated from your datasets.

7.2 Model-comparison table

Report the metrics from the held-out test set. The following is a blank template, not experimental results.

Disease	Dataset size	Selected model	Accuracy	Precision	Recall	F1-score
Diabetes prediction	768	XGBoost	74.03%	63.46%	61.11%	62.26%
Heart disease prediction	303	Random Forest	90.16%	86.67%	92.86%	89.66%
Chronic Kidney Disease (CKD) prediction	400	Random Forest	98.75%	98.04%	100.00%	99.01%
Thyroid cancer recurrence prediction	383	Random Forest	96.10%	100.00%	86.36%	92.68%

Repeat for each disease actually supported by your code. Do not include a model for a disease if it was not trained and evaluated for that disease.

7.3 Additional performance comparison

Model	Mean F1 ± SD	Training time (s)	Mean inference latency (ms)	Artifact size (MB)
Logistic Regression	Measure	Measure	Measure	Measure
Decision Tree	Measure	Measure	Measure	Measure
Random Forest	Measure	Measure	Measure	Measure
XGBoost	Measure	Measure	Measure	Measure
Gaussian Naive Bayes	Measure	Measure	Measure	Measure

Measure these properties using the same hardware and a consistent evaluation procedure. Record the number of repeated runs used for the stability calculation.

7.4 ROC-AUC comparison of selected models

Figure 2 compares the ROC-AUC values of the selected models across the four conditions. The highest reported ROC-AUC values are 83.00% for diabetes, 96.00% for heart disease, 100.00% for CKD, and 99.50% for thyroid cancer recurrence.

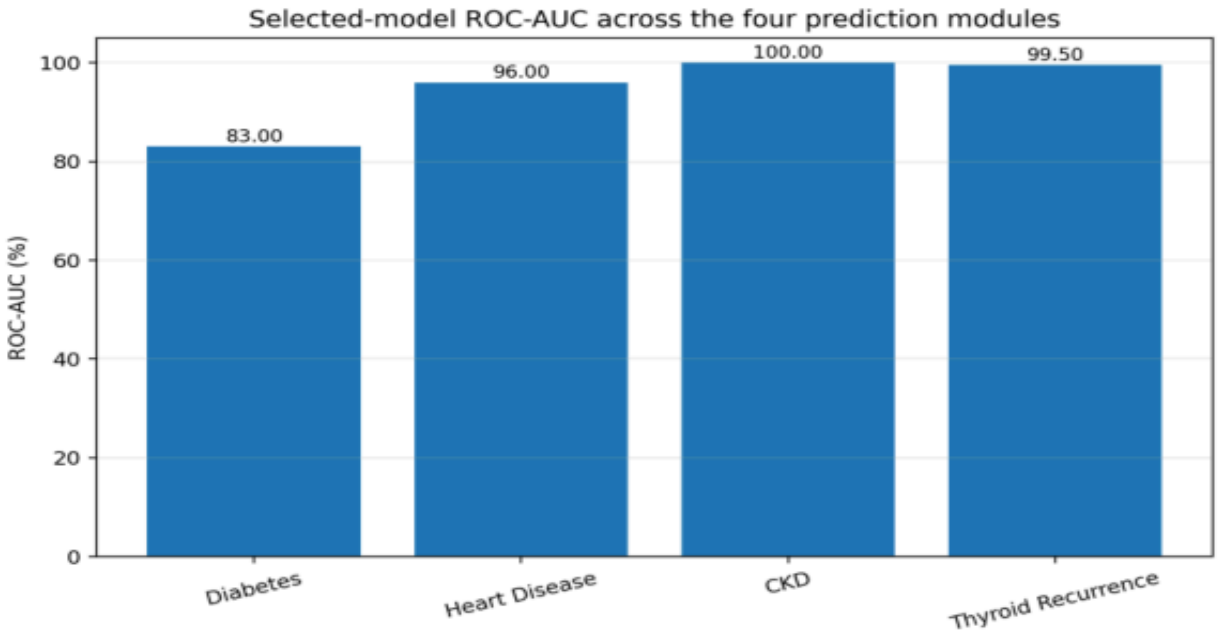


Fig. 2. ROC-AUC of the selected models across the four conditions.

Figure 2. ROC-AUC of the selected models across the four conditions.

7.5 Confusion-matrix analysis

The confusion matrices show the classification outcomes for the selected model in each disease module. Diabetes (XGBoost) produced 81 true negatives, 19 false positives, 21 false negatives, and 33 true positives. Heart disease (Random Forest) produced 29 true negatives, 4 false positives, 2 false negatives, and 26 true positives. CKD (Random Forest) produced 29 true negatives, 1 false positive, 0 false negatives, and 50 true positives. Thyroid recurrence (Random Forest) produced 55 true negatives, 0 false positives, 3 false negatives, and 19 true positives.

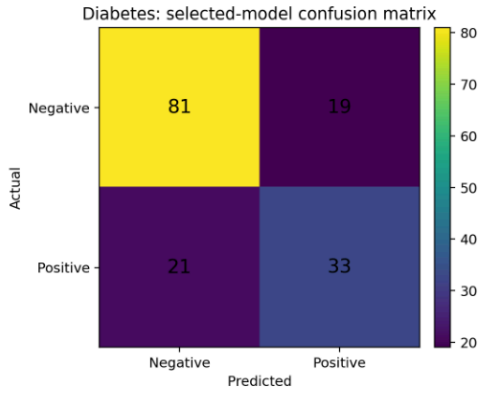


Fig. 3. Diabetes confusion matrix for selected XGBoost.

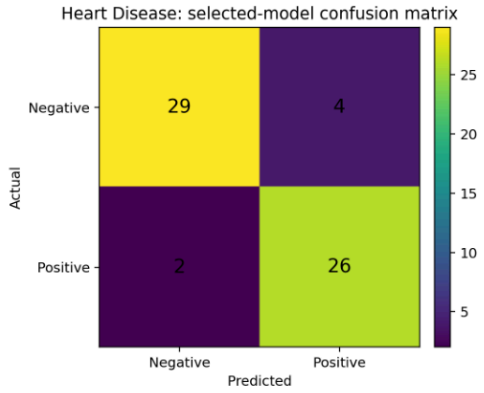


Fig. 4. Heart disease confusion matrix for selected random forest.

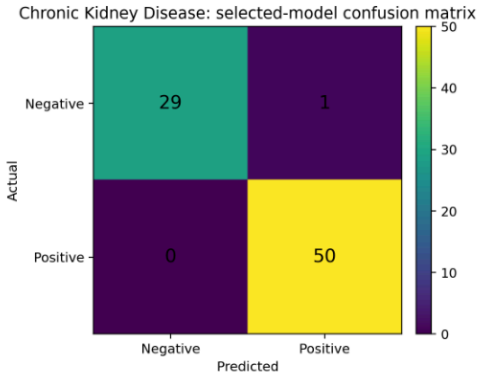


Fig. 5. CKD confusion matrix for selected random forest.

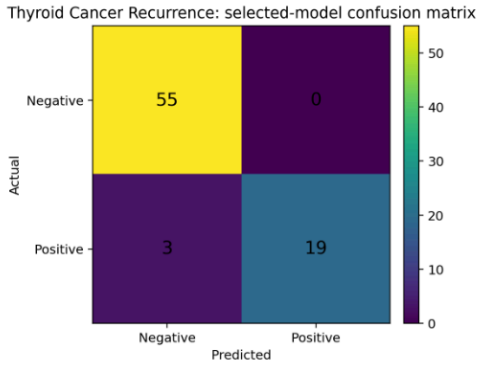


Fig. 6. Thyroid recurrence confusion matrix for selected random forest.

does not prove high concurrent throughput, and strong perfor-

Figures 3–6. Confusion matrices for the selected diabetes, heart disease, CKD, and thyroid recurrence models.

7.6 Comparison of candidate machine-learning models

Figures 7–10 compare Logistic Regression, Decision Tree, Random Forest, XGBoost, and Gaussian Naive Bayes using accuracy, precision, recall, F1-score, and ROC-AUC. For diabetes, Random Forest has the highest accuracy (76.62%), while XGBoost has the highest ROC-AUC (83.00%) and is selected. For heart disease, Random Forest achieves the highest accuracy (90.16%) and ROC-AUC (96.00%). For CKD, XGBoost reaches 100.00% across all five metrics, while Random Forest also has 100.00% ROC-AUC. For thyroid recurrence, Random Forest has the highest ROC-AUC (99.50%), although Logistic Regression and XGBoost have higher accuracy (97.40%).

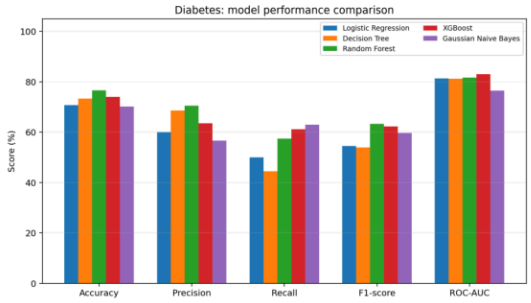


Fig. 7. Diabetes model comparison.

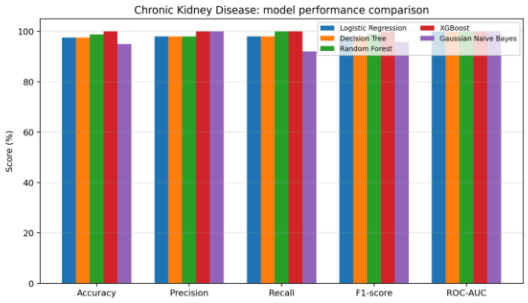


Fig. 9. CKD model comparison.

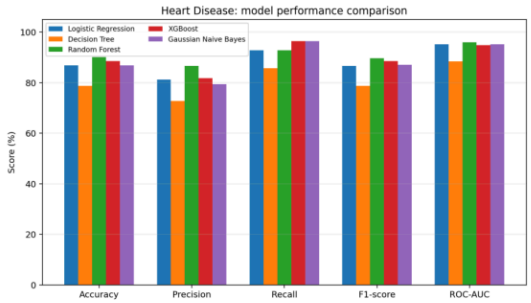


Fig. 8. Heart disease model comparison.

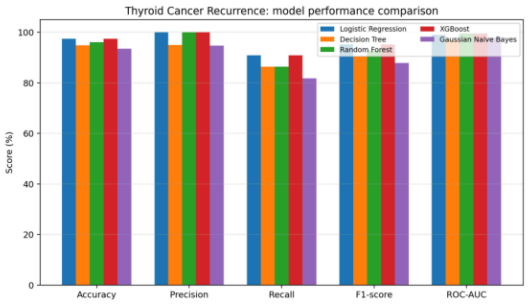


Fig. 10. Thyroid recurrence model comparison.

A lower standard deviation indicates lower observed variation under the chosen resampling protocol, but it does not guarantee external validity. The same seeds and split definitions should be applied to every candidate model so the comparison is paired. The current stored experiment contains one split only; therefore, this paper does not claim a measured stability ranking. The repeated-seed experiment must be run before



Figures 7–10. Candidate-model performance comparisons for diabetes, heart disease, CKD, and thyroid recurrence.

7.7 Discussion and limitations

The results show that the best-performing classifier differs by disease, supporting disease-specific model selection rather than using one classifier for every task. However, the CKD and thyroid results are unusually high and require additional checks for duplicate records, target leakage, preprocessing errors, and overlap between training and test data. The evaluation uses one held-out split and no independent external clinical cohort. Therefore, the results are preliminary and should not be interpreted as evidence of clinical validation. Repeated-split stability, controlled execution-time measurements, memory profiling, and scalability testing are not reported as completed experiments in the paper.

8. CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

EarlyMed-AI presents a modular approach to healthcare analytics by integrating disease-specific machine-learning models, comparative evaluation, explainability, and a web-based interface. The framework is designed to make model-generated disease-risk estimates more understandable through SHAP and LIME explanations.

The project also emphasizes evaluating more than predictive accuracy. Stability, execution time, model storage requirements, and scalability are important when considering the practical deployment of machine-learning systems.

The current prototype should be regarded as a research and educational decision-support framework. Its predictive performance, reliability, and clinical utility must be demonstrated through properly designed experiments and validation before any clinical use. The final conclusion should identify the best-performing model for each disease only after the actual test results have been calculated.

8.2 Future scope

1. Larger and more diverse datasets: Evaluate generalization across hospitals, age groups, and population groups.
2. External validation: Test models on datasets independent of those used for training.
3. Probability calibration: Assess whether predicted risks correspond to observed event frequencies.
4. Improved missing-data handling: Evaluate alternative imputation approaches and make missingness more transparent to users.
5. Advanced models: Investigate suitable gradient-boosting and neural-network methods when sufficient data are available.
6. Explainability stability: Study how SHAP and LIME explanations change under small input perturbations and different background samples.
7. Scalable deployment: Evaluate concurrent requests, memory consumption, and application latency.
8. Privacy and security: Add appropriate access control, secure data handling, retention policies, and protection for sensitive medical information.

9. Fairness evaluation: Measure performance across relevant demographic subgroups where the data and governance arrangements permit.
10. Clinical validation: Conduct prospective or external validation studies with appropriate clinical oversight before considering deployment.

8.3 SDG contribution

SDG	Project relevance
SDG 3 — Good Health and Well-Being	Research into early risk estimation and preventive healthcare support
SDG 9 — Industry, Innovation and Infrastructure	Application of machine learning and explainability to a healthcare technology prototype
SDG 10 — Reduced Inequalities	Future opportunity to evaluate model performance across demographic groups and identify disparities

These are intended contributions. The project should not claim that it has improved population health or reduced inequality without supporting evidence.

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